Fizika

9 sinf

MAVZU: Izojarayonlar.

Izotermik jarayon

Izojarayonlar

Berilgan gazning bitta parametri o‘zgarmas bo‘lganda qolganlari orasidagi bog‘lanishini tavsiflaydigan jarayon ***izojarayon*** deb ataladi.



Izotermik jarayon

m=const va T=const

$$p_1V_1 = \frac{m}{M}RT \quad \text{va} \quad p_2V_2 = \frac{m}{M}RT$$

$$\boldsymbol{p_1V_1 = p_2V_2} \qquad \textbf{pV= const}$$

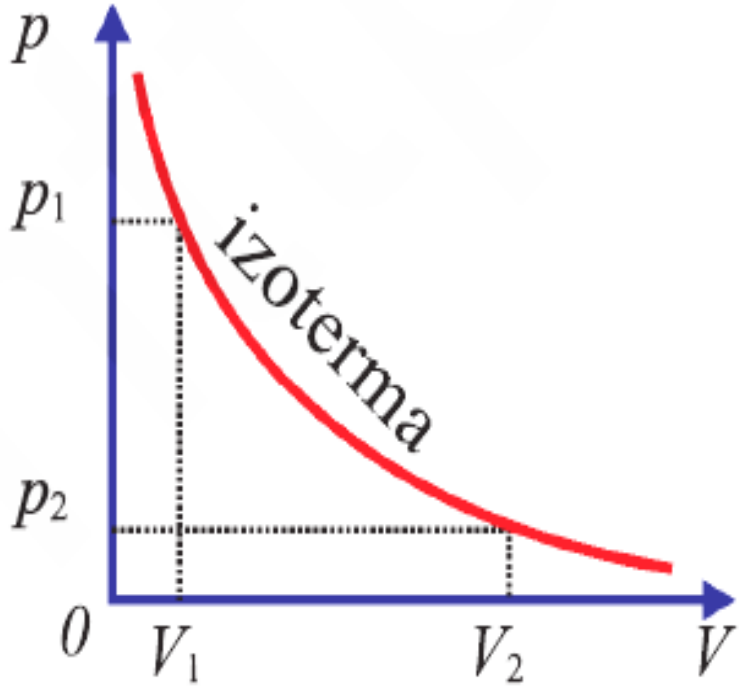
Izotermik jarayon 1662-yil R.Boyl va 1676-yil E.Mariott tomonidan kashf etilgan. Shuning uchun Boyl- Mariott qonuni deyiladi.

Izotermik jarayon

T=const

pV=const

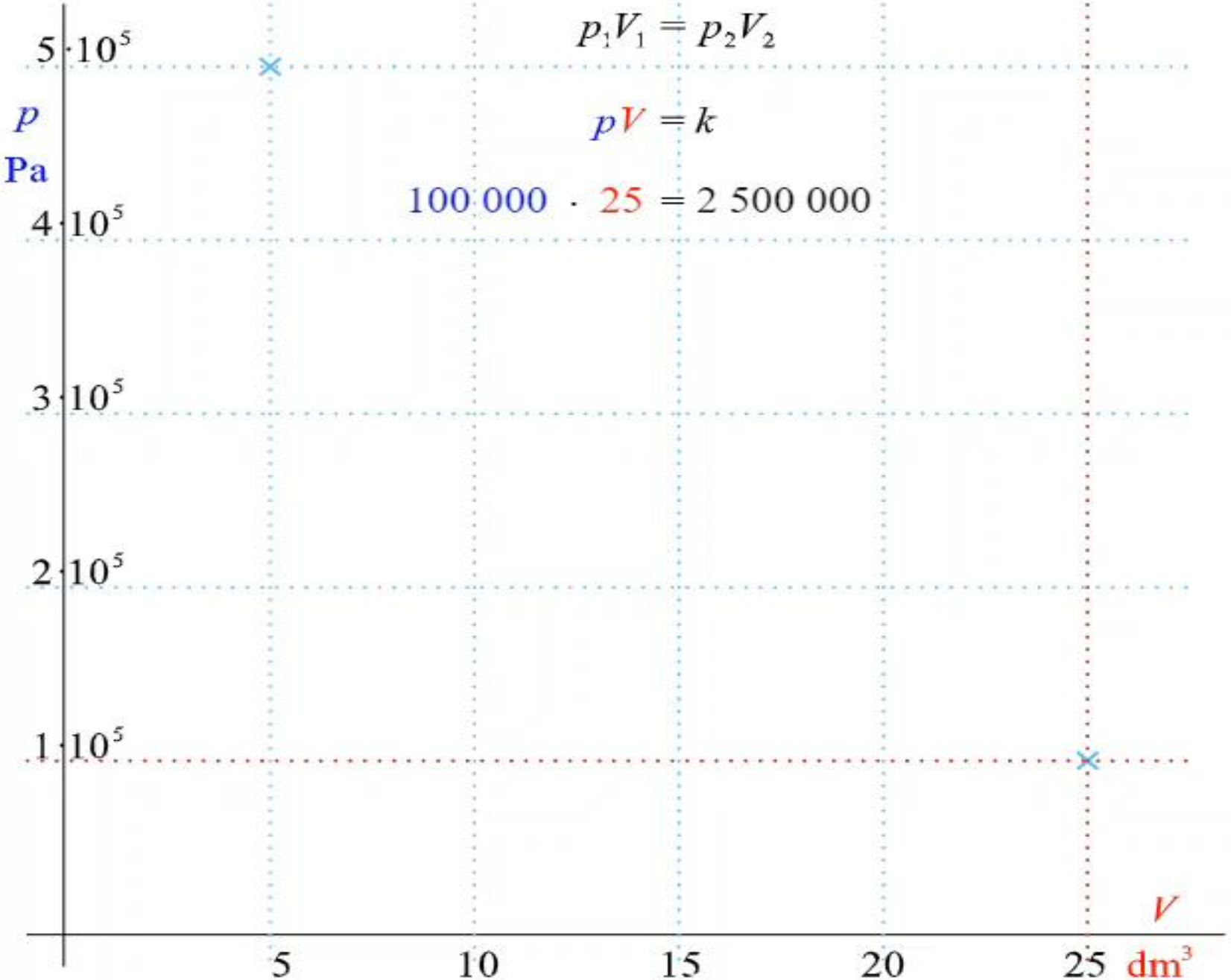
Temperatura o‘zgarmas bo‘lganda gaz bosimining hajmga bog‘lanish chizig‘i (giperbola) izoterma chizig‘i deyiladi. $p \sim 1/V$

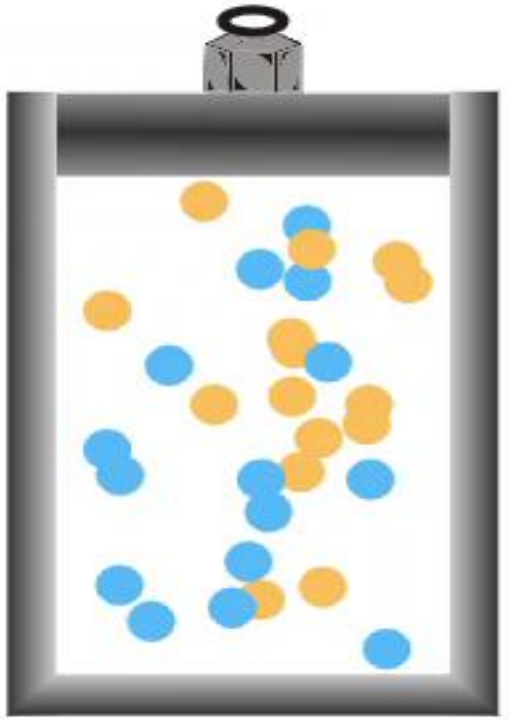


$p_1V_1 = p_2V_2$

$pV = k$

$100\,000 \cdot 25 = 2\,500\,000$





$Q_T = W'$

$\Delta U = 0$



O‘zgarmas temperaturada berilgan gazning bosimi hajmiga teskari proporsional ravishda o‘zgaradi.

$$\rho_1 = \frac{m}{V_1} \qquad \text{va} \qquad \rho_2 = \frac{m}{V_2}$$

Bu ifodalarni bir-biriga nisbatini olsak, quyidagi ifoda hosil bo‘ladi:

$$\frac{\rho_1}{\rho_2} = \frac{V_2}{V_1} = \frac{P_1}{P_2}$$

8 – mashq

1- masala

Gazning dastlabki hajmi 0,2 l , bosimi esa 300 kPa bo‘lgan. Gaz izotermik kengayib, bosimi 120 kPa ga erishdi. Gazning keyingi hajmini toping.

Berilgan:

$V_1 = 0{,}2\ l$

$p_1 = 300\text{kPa}$

$p_2 = 120\text{kPa}$

$T = \text{const}$

Formulasi:

$$p_1 V_1 = p_2 V_2$$

$$V_2 = \frac{p_1 V_1}{p_2}$$

Topish kerak: V_2-?

Yechish:

$$V_2 = \frac{300\ \text{kPa} \cdot 0{,}2\ l}{120\ \text{kPa}} = 0{,}5\ l$$

Javob: $V_2 = 0{,}5\ l$

8- mashq

3-masala.

Normal atmosfera bosimi sharoitida ideal gaz 50 l hajmni egallaydi. Agar bosim 4 marta ortsa, gaz qancha hajmni egallaydi? Temperatura o‘zgarmas.

Berilgan:

$T = const$

$V_1 = 50\ l$

$p_2 = 4p_1$

Topish kerak: $V_2 - ?$

Formula:

$p_1V_1 = p_2V_2$

$$V_2 = \frac{P_1V_1}{p_2} = \frac{p_1V_1}{4P_1} = \frac{V_1}{4}$$

Yechish: $V_2 = \frac{50\ l}{4} = 12{,}5\ l$

Javob: $V_2 = 12{,}5\ l$

19-test 45- bet

Izotermik jarayonda gaz bosimi 4 marta ortdi. Bunda gaz konsentratsiyasi qanday o‘zgaradi?

A) 2 marta ortadi B) 4 marta ortadi

C) 4 marta kamayadi D) 2 marta kamayadi

$\frac{p_2}{p_1} = 4$ $T_1 = T_2 = T$

Yechish: $p = nkT$ $\begin{cases} p_1 = n_1 kT \\ p_2 = n_2 kT \end{cases} \Rightarrow \frac{p_1}{p_2} = \frac{n_1}{n_2} \Rightarrow \frac{p_2}{p_1} = \frac{n_2}{n_1}$

$\frac{n_2}{n_1} = 4$

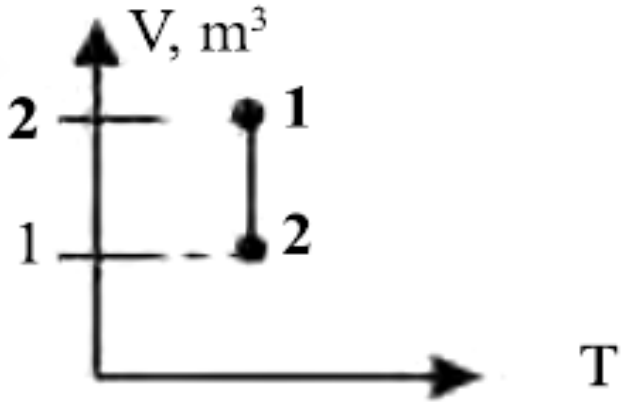
Javob: B) 4 marta ortadi

20-test 45- bet

Rasmda tasvirlanganidek gaz 1-holatdan 2-holatga o‘tganda, uning bosimi qanday o‘zgaradi?

A) 4 marta ortadi B) 4 marta kamayadi

C) o‘zgarmaydi D) 2 marta ortadi



$T = const$; $\frac{V_1}{V_2} = 2$

Yechish: $p_1V_1 = p_2V_2$

$\frac{V_1}{V_2} = \frac{p_2}{p_1} = 2$

Javob: D) 2 marta ortadi.

Mustaqil bajarish uchun topshiriqlar

1. Izojarayonlar deb qanday jarayonga aytiladi?
2. Qanday jarayon izotermik jarayon deb ataladi?
3. Izoterma chizig‘i nima va u qanday chiziqdan iborat?
4. Izotermik jarayonda gazning zichligi hajmga qanday bog‘langan?
5. 8-mashqning 2- va 4- masalalarini yechish.